

Thermo-mechanical Impact of Thruster Plume Impingement on Space Debris

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An analytical model of the impact of plume impingement from a reaction control thruster onto a debris surface has been created. This scenario is relevant to the study of possible ways to detumble space debris objects prior to their capture in active debris removal missions. Simons' plume model has been used to estimate the local density, velocity, and temperature in the plume. A distance of 10m between the thruster and debris object has been assumed, and at this distance the flow densities are of the order of 10^{-8} kg/m^3 for a 5N hydrazine monopropellant thruster. The plume was found to behave as a free-molecular flow at this distance. Subsequent analysis of the plume impingement onto a glass surface layer of a solar panel on the debris object, found peak heat fluxes of approximately 150 W/m^2 , causing a temperature increase up to 14 K during an example detumbling sequence. Peak stresses on the surface were found to be on the order of 1 MPa . Comparison of these results to failure limits related to adhesive melting and fracture propagation found the risk of debris breakup to be low. However, there is a large degree of uncertainty in the analytical model presented, and it is suggested that future work should focus on comparing these results to those generated by a numerical method, in particular the direct simulation monte carlo method.

Nomenclature

A	=	Area, m^2
A_P	=	Plume constant
B	=	Molecular rotational constant
a	=	Initial microcrack length, m
C	=	Arbitrary constant
C_{TE}	=	Coefficient of thermal expansion
c_p	=	pressure coefficient
c_{rho}	=	angular decay constant
d	=	Distance from nozzle exit plane to debris surface, m
F	=	Force, N
G_c	=	Fracture energy, J/m^2
K_c	=	Fracture Toughness, $\text{MPa}\sqrt{m}$
Kn	=	Knudsen Number
k_B	=	Boltzmann constant
L	=	Characteristic Length, m
M	=	Mach Number
m	=	Mass, kg

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P	=	Bird's parameter
p	=	Pressure, Pa
Q	=	Heat energy, J
R	=	Universal gas constant, $J/molK$
Re	=	Reynolds Number
r	=	Radial Distance from centre of nozzle exit plane, m
S	=	Speed Ratio
T	=	Temperature, K
u	=	Velocity, m/s
$\bar{u}$	=	Average velocity, m/s
$\mathcal{R}$	=	Specific gas constant, J/kgK
$\dot{q}$	=	Heat flux, J/kgK
σ_T	=	Collision cross section, m^2
θ	=	Angle of Incidence, $^\circ$
ϵ	=	emissivity
Θ	=	Boundary-layer momentum thickness, m
ρ	=	Density, kg/m^3
τ	=	Shear stress, Pa
ν	=	Collision frequency, $1/s$
ϕ	=	Angle between plume centreline and streamline, rad
σ	=	Momentum accommodation coefficient
σ_B	=	Stefan-Boltzmann constant
α	=	Thermal accommodation coefficient
γ	=	Ratio of specific heats
λ	=	Mean free molecular path, m
ω_e	=	fundamental molecular vibrational frequency Hz
Subscripts		
E	=	Nozzle exit flow conditions
0	=	Stagnation flow conditions
n	=	Normal flow component
t	=	Tangential flow conditions
w	=	Impinged surface conditions
lim	=	Limiting value
m	=	focus node during thermal solver time step
n	=	adjacent node during thermal solver time step
Superscripts		
j	=	time step in thermal solver
θ	=	evaluated for the current incidence angle
$*$	=	Throat flow conditions

I. Introduction

The accumulation of space debris presents a direct challenge to the sustainability of orbital operations and the long-term viability of space exploration. Space debris comprises non-functional satellites, spent rocket stages, and fragmentation products from orbital collisions or explosions. Debris typically travels at velocities exceeding 7 km/s in low Earth orbit (LEO). At such speeds, even millimetre-sized particles pose a significant risk to spacecraft operations; larger debris is capable of causing catastrophic collisions that generate large amounts of fragmentation debris [1]. As of 2023, the European Space Agency (ESA) estimates that approximately 40,500 objects larger than 10 cm, 1.1 million objects between 1 cm and 10 cm, and over 130 million objects smaller than 1 cm are currently orbiting Earth [2]. The most congested regions are LEO (below 2,000 km) and geostationary orbit (GEO, approximately 35,786 km), where there is a particular need for enhanced tracking, collision avoidance, and debris mitigation strategies such as active debris removal (ADR). Active debris removal (ADR) is defined as the capability to interact with passive orbiting objects, in order to

reduce their orbital lifetime [3]. It is an important concept for space sustainability, and has been shown to be necessary to control the unstable growth rate of debris in low Earth orbit (LEO) [4, 5]. Several studies have assessed the technical and economic feasibility of ADR [6, 7], and the last decade has seen multiple ADR demonstration missions.

Prior to such missions, autonomous rendezvous and docking capabilities were demonstrated through on-orbit servicing missions such as JAXA's ETS-VII in 1997 ([8]), with the knowledge gained proving essential to the development of today's ADR concepts [9]. ClearSpace-1, a mission led by ClearSpace under ESA contract, aims to be the first to capture and de-orbit a real piece of space debris: a defunct Vespa upper stage [10]. Other missions such as Surrey Space Centre's RemoveDEBRIS mission (2018), and Astroscale's ELSA-d mission (2021), successfully deployed and then captured simulated debris objects [11, 12]. A further mission by Astroscale in 2024 (ADRAS-J), focused on proximity operation testing; it approached to within 15m of a debris object [13]. This is the closest approach to-date by a commercial satellite to a debris object. Unlike on-orbit servicing missions, the targets of ADR missions, are by definition uncooperative. This means that additional care must be taken during any close proximity operations, to minimise the risk of collision with the debris [14]. Applicable standards for the mitigation of space debris are NASA-STD-8719.14C in North America ([15]), and ISO 24113 in Europe (after adoption by ECSS) [16]. Both of these standards state any debris greater than 10cm poses a serious risk of damage, were it to be involved in a collision. For this reason, companies such as Astroscale have adopted a zero-debris policy during ADR missions, aiming to create no new debris during operations.

One way of minimising the risk of debris creation is to reduce the spin rate differential between a debris object and the chaser spacecraft [7]. Nishida suggests that a spin rate differential less than $3^\circ/s$ is necessary to easily capture a debris object using a robotic arm [17]. Indeed, it has become generally accepted that the ability to detumble debris is necessary for any rigid-connection ADR concept such as robotic arm grappling [18]. Several methods for detumbling space debris have been proposed: nets and tethers [19]; laser ablation [20]; flexible brushes [21]; eddy currents [22]; and gas thruster plume impingement [23–25].

It is this final method that is the focus of this paper. Nakajima highlights the favourable properties of plume impingement as a detumbling method, namely that no physical contact between debris and chaser satellite is needed during detumbling [24]. Further, a chaser satellite will generally already possess the capability to perform plume impingement, by use of its reaction control system (RCS) thrusters. Figure 1 demonstrates how these thrusters may be used to impart a detumbling moment onto a debris object.

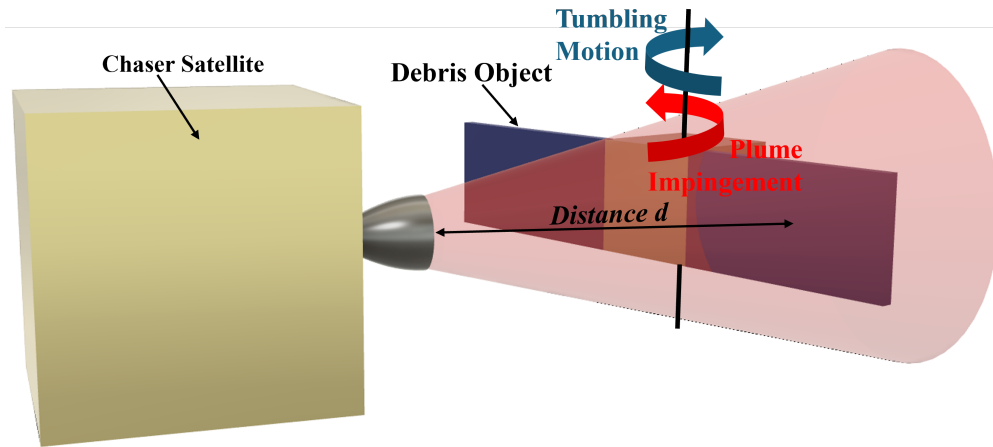


Fig. 1 Plume impingement concept for detumbling a debris object.

Plume impingement as a method of detumbling has been studied in-depth as the COBRA concept, developed as part of ESA's SysNova technology assessment scheme [25]. The COBRA concept looked specifically at detumbling ENVISAT, a large Earth-Observation satellite, using a 5N thruster. ENVISAT has been the focus of similar ADR studies by Ferrari and Vetrivano [23, 26]. Its large size, large mass, and common (sun-synchronous) orbit, greatly increases its associated risk. Indeed, it has been identified that almost all major spacecraft targets for ADR lie in the region of sun-synchronous

orbits: inclinations of 95-100°, and altitudes of 500-1000km [27]. Servadio et al. determined ten priority targets for ADR in this orbital region, following the creation of a risk index based on the debris' mass and residual lifetime, as well as the background density across its orbit [28]. All ten of the targets identified were defunct US satellites launched between 1964-1972. The characteristic lengths of these satellites range from 1.2-4.2m, and the dry masses from 50-192kg. For this reason, this work focuses on satellites of this scale, and it has been assumed that the general shape of a target debris object resembles Figure 1- specifically a central 'box' with deployed solar panels.

Most studies on debris detumbling via plume impingement focus on the scenario from a guidance, navigation, and control (GNC) standpoint. That is, they concentrate on the force and time required to generate sufficient torque to detumble a debris object. The aim of this work was instead to determine whether there is a significant risk of debris breakup during a detumbling sequence. This was achieved by analysing the combined thermal and mechanical effects of a thruster plume for two specific debris breakup scenarios: the impingement of an embrittled, micro-cracked solar panel; and impingement effects on an adhesive layer of a solar panel.

Section II presents a background of methods commonly used to analyse plume impingement scenarios, in particular analytical techniques. Section III presents the methodology used in this work to model the plume, its impingement, and the failure mechanics considered when determining the risk of debris breakup. Section IV presents the results of this work, and are discussed along with an evaluation of the methodology in Section V. The findings are concluded and recommendations for future work suggested in section VII.

II. Background

Dettleff presents a comprehensive review of techniques to study plume flows and plume impingement [29]. In it he explores different methodologies and their application in different regions of a plume flow. A key concept of plume flows in space, is the effect of large density variations: from relatively high densities at the exit of the nozzle, to extremely low densities in the far field vacuum. The collision frequency ν between particles in the plume is proportional to the density, and hence far away from the nozzle exit there becomes insufficient molecular collisions to maintain equilibrium within the flow. With this comes the breakdown of continuum models characterised by the Navier-Stokes equations. That is, there are insufficient particular collisions to transfer kinetic energy, and the adiabatic relation $T \propto P^{((1-\gamma)/\gamma)}$ no longer holds. Instead, the flow begins to behave as free molecules, described by the Maxwell-Boltzmann equation of kinetic theory [30]. The transition of a flow from 'continuum' to 'free molecular' behaviour is commonly described by the Knudsen number, defined as the mean free path normalised by a characteristic length: $Kn = \lambda/L$. The mean free path of a particle is the average distance between travelled by a particle between two collisions, defined mathematically as:

$$\lambda = u/\nu = \frac{m}{\sqrt{2}\rho\sigma_T} \quad (1)$$

It is appropriate to note here that all parameters used throughout this paper, and their subscripts, are defined in the nomenclature section; definitions are only reiterated in the report's body where necessary for clarity. Figure 2 shows the different regions of a flow as defined by Knudsen number, and the typical values of Kn for which each region is valid.

The choice of characteristic length used to define Kn differs across plume studies. In plume impingement studies, where the focus is on gas-surface interactions, it is more common to use the length from the nozzle exit to the impinging surface or a characteristic length of the surface itself. This way, the choice of L allows the impingement model to reflect the equations of motion applicable over the length scale of the impinged surface (i.e. continuum and Navier-Stokes, or free-molecular and kinetic theory). Kn then, is particularly suited to defining the nature in which a gas plume will impinge onto a surface. Bird introduced another parameter that more explicitly describes the breakdown of equilibrium within a flow (rarefaction), that is not subject to the discrepancies introduced by needing to choose a normalisation factor. Bird's Parameter is defined as

$$P = \frac{u}{\nu\rho} \cdot \left| \frac{d\rho}{dr} \right| \quad (2)$$

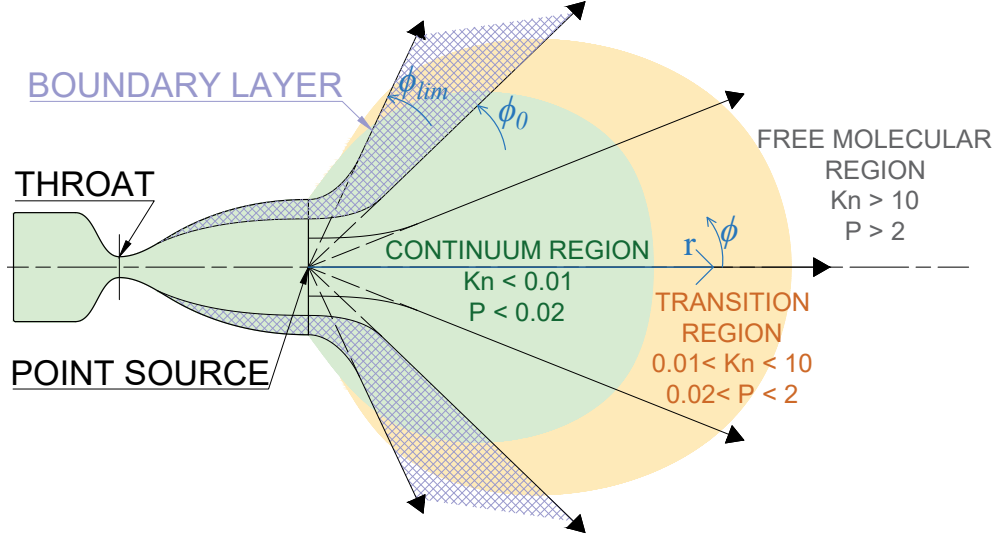


Fig. 2 Plume flow regions, and their characteristic Knudsen and Bird parameter ranges.

Specifically, it describes the onset of thermal non-equilibrium of the flow, whereby the plume temperature plateaus to a minimum, even as density continues to decay. This 'freezing' of the flow more closely describes its rarefaction, and is directly related to the breakdown of continuum assumptions. As such, many models use P to define the limits of where continuum and free-molecular *plume* models are used, and Kn to define where different *impingement* models are used.

The Direct Simulation Monte Carlo (DSMC) method is accepted as the best current method to describe the transition and free molecular plume regions. DSMC is a phenomenological method that models individual particular collisions, rather than assuming continuous distribution functions of velocity [31]. This, unsurprisingly, can result in high computational cost if the number of particular collisions is high [32].

There are also several difficulties associated with experimental techniques of measuring plume impingement, primarily due to the difficulty of maintaining realistic orbital vacuum levels whilst firing a thruster [33]. This limits the mass flux that a thruster can eject during testing. This limit is seen in many experimental results reported in literature, which use small RCS thrusters, in particular the ERNO 0.5N thruster [34–37]. Further difficulties arise when studying hydrazine thrusters. The dangers of hydrazine are perhaps most aptly described by Clark who remarked its ability to "eat through everything - metal, rubber, glass, and even the engineer who has to work with it" [38].

Luckily, in his review, Dettleff states that it is possible to describe a plume flow to a sufficiently exact degree using analytical methods. Such analytical methods are particularly applicable for hypersonic, vacuum situations. Commonly in analytical methods, the flow is modelled to expand radially with straight streamlines from a point source at the centre of the exit plane [39]. In these scenarios, the expansion of the plume at the nozzle exit is large in angle and Mach number, such that the flow appears to expand radially with straight streamlines from a point source at the centre of the exit plane [39]. From this assumption, the decay of density of the plume is characterised in its most basic formulation as:

$$\rho(r, \phi) = C \frac{1}{r^2} f(\phi) \quad (3)$$

It can be seen that the density decays both with radial distance from the nozzle, and also with angle from the centreline. In most cases the plume can be assumed axisymmetric about the nozzle centreline, and hence a 2D plume model is sufficient. Peters and Olmos identified two popular analytical models used for plume impingement modelling: Simons' model and Fehse's model [25, 40, 41]. Both take the form of Eq. 3. Fehse's model describes pressure distribution, and was used by Ferrari et al when analysing the detumbling of ENVISAT [23]. Simon's model has been used more in plume impingement studies, in part due to its ability to model the viscous effects in boundary layer region of the

plume. Simons' defines two important angles for the flow: θ_{lim} the most extreme angle of the flow, where the velocity approaches zero; and θ_0 , which is the limit of the isentropic core. Figure 2 shows these two angles. A further extension to Simons' model was presented by Legge and Boettcher, primarily aiming to better capture the effects of rarefaction within the boundary layer of the plume [37]. A freezing surface at $P = 2$ was used to define the onset of the free molecular region. It is assumed that no molecular collisions occur downstream of this freezing surface, and hence local flow velocity and temperature hit maximum and minimum values respectively.

In the same work, Legge and Boettcher use the method of Schaaf and Chambre to describe impingement of the plume onto a surface in the free molecular region. Schaaf and Chambre presented the use of three accommodation coefficients to describe the degree to which the interaction of particles with a surface is specular or diffuse [42]: thermal accommodation coefficient α_e to describe energy transfer; and normal and tangential momentum accommodation coefficients α_n and α_t . Figure 3 shows the distinction between specular and diffuse collisions. In specular reflection, there is no momentum or energy transfer from the particle to the surface. In fully diffuse interactions, the incident molecule is brought to equilibrium state with the surface.

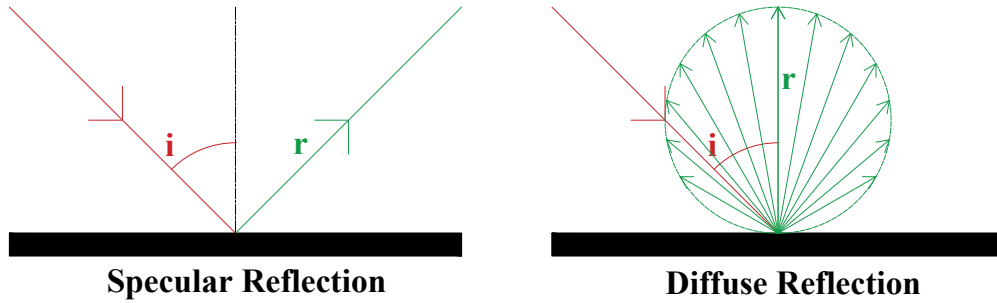


Fig. 3 Specular and diffusive gas-surface interactions.

III. Methodology

Table 2 and Fig. 4 show the flow properties and nozzle dimensions of the generalised thruster considered, as suggested by Astroscale. A bell-shaped nozzle has been chosen, where the angle between the wall tangent and the nozzle centreline is zero at the exit plane. This has the effect of narrowing the thruster plume, compared to a similar conical nozzle [43].

Table 2 Nozzle flow properties.

Parameter	Value
Thrust [N]	5
Propellant Type [-]	hot-gas monopropellant
Propellant Species [-]	N_2H_4
Chamber Pressure p_0 [bar]	11
Chamber Temperature T_0 [K]	1200
Expansion Ratio [-]	121

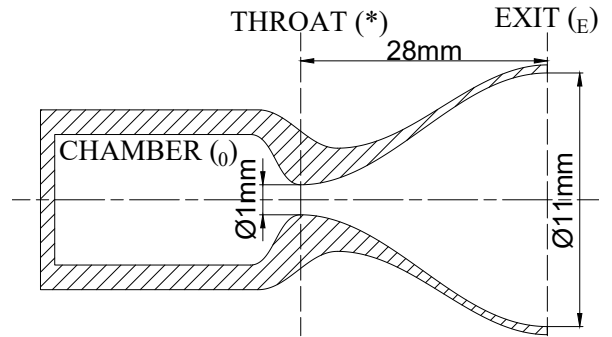


Fig. 4 Nozzle dimensional parameters.

Monopropellant hydrazine thrusters have been the focus of multiple plume impingement studies, in particular Boyd's work [34, 35]. Like Boyd, this analysis used Simons' method to model the plume flow, as it was presented and extended by Legge and Boettcher [37]. NASA's *Chemical Equilibrium with Applications* (CEA) code, was used to determine

the input parameters needed for Simons' model [44]. The inputs to CEA itself are those of Table 2. The outputs of Simons' model were used to determine the energy and momentum transfer onto the impinging debris surface. Schaaf and Chambre's method for determining energy and momentum transfer in a free molecular flow has been used to model the plume impingement [42]. The calculated impingement effects were then used to determine peak stresses and maximum temperature increases during an example detumbling sequence. From these results, the risk of debris breakup for two specific scenarios could be assessed; fracture propagation on a micro-cracked solar panel glass layer; and the melting of an adhesive layer of a solar panel. The general workflow of this methodology is shown in Fig. 5.

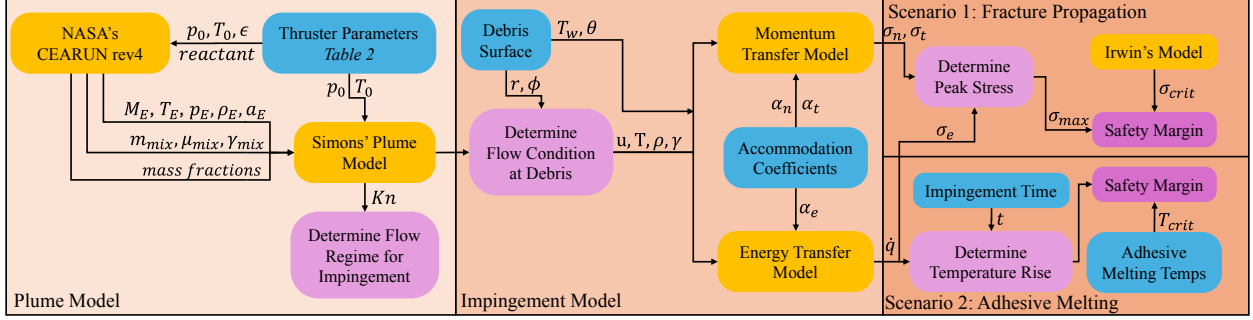


Fig. 5 General method workflow. Yellow stages are methods, blue are inputs, and purple are key results.

III.i. Determining Nozzle Exit Plane Properties

Simons' model requires, as inputs, flow properties at the exit plane of the nozzle; Legge notes the particular importance of knowing γ accurately at this location [37]. To determine the relevant parameters, the *rocket* solver of *CEARUN rev4* was used [44]. This program computes throat and exit plane parameters, and takes as input: chamber pressure; reactant species; reactant temperature; and nozzle expansion ratio. Notably, CEA does not allow chamber temperature to be specified, and instead a reactant temperature of 579.35K was specified to achieve the desired chamber temperature $T_0 = 1200K$. CEA defines the chamber conditions as those where velocity is zero. Hence, 'chamber' and 'stagnation' conditions are identical here. The run file of the CEA analysis can be found in the Appendix. The reactant species chosen for the CEA analysis was liquid Hydrazine " $N_2H_4(L)$ " (as opposed to the gaseous form), as it is the state most commonly found in monopropellant RCS subsystems used on spacecraft to date [45, 46].

The CEA *rocket* solver first models the chemical reaction of the chamber, and subsequently models the isentropic expansion of the reacted gas mixture through a nozzle. The chemical reaction is computed by minimising the Gibbs energy, and the isentropic expansion is calculated according to isentropic flow relations, found in common aerodynamics references such as Anderson [47]. The *rocket* solver can be run in two modes: frozen flow, where the chemical composition is fixed after combustion; or a shifting equilibrium flow, where the flow properties are re-solved for equilibrium throughout the nozzle length. In this analysis, a shifting-equilibrium model was used, in order to capture some of the condensation and recombination effects known to occur in the nozzle of small thrusters [48]. The better estimation of product species mass fractions also aided in subsequent analysis of Knudsen number. Both the chemical reaction, and isentropic expansion solver introduce assumptions about the nozzle flow:

- complete combustion occurs in the chamber
- thermodynamic equilibrium is achieved at all points in the combustion chamber and nozzle
- the flow is a steady with time, and an ideal gas
- the flow expands isentropically with no viscous boundary layer
- no shockwaves form in the nozzle

III.ii. Plume Model

As stated, Simons' method was used for modelling the thruster plume. This model was chosen firstly because its analytical nature leads to small computational cost, compared to numerical methods such as MOC and DSMC. Secondly, Peters and Olmos showed that Simon's model aligned more closely with DSMC solutions, compared with Fehse's analytical model [25]. Simons' model describes the decay of density in the plume:

$$\rho(r, \phi) = \rho(r, \phi = 0) \cdot f(\phi) = \rho^* A_P \left(\frac{r^*}{r} \right)^2 \cdot f(\phi) \quad (4)$$

The angular decay function is defined differently for the isentropic plume core ($\phi \leq \phi_0$), compared to the viscous boundary layer ($\phi_0 < \phi \leq \phi_{lim}$):

$$f(\phi) = \begin{cases} \cos^{\frac{2}{\gamma-1}} \left(\frac{\pi \phi_0}{2 \phi_{lim}} \right), & \phi \leq \phi_0 \\ f(\phi = \phi_0) e^{-c_\rho (\phi - \phi_0)}, & \phi_0 < \phi \leq \phi_{lim} \end{cases} \quad (5)$$

ϕ_{lim} and ϕ_0 are calculated using the Prandtl-Mayer angle and boundary layer thickness δ_E , evaluated at the nozzle exit. Since CEA assumes inviscid flow, δ_E was estimated from Ref. [37]. c_ρ and A_P are plume constants affecting angular and radial decay respectively, and are defined in Eqs. 6 and 7. $\bar{u}_{lim}$ was approximated as $\bar{u}_{lim} = 0.75 u_{lim}$.

$$c_\rho = A_P \left(\frac{\gamma + 1}{\gamma - 1} \right)^{1/2} \frac{2 \bar{u}_{lim}}{u_{lim}} \left(\frac{r_E}{2 \delta_E} \right)^{\frac{\gamma-1}{\gamma+1}} \quad (6)$$

$$A_P = \frac{u^*}{2 u_{lim} \int_0^{\phi_{lim}} f(\phi) \sin \phi d\phi} \quad (7)$$

These equations allow the density field of the plume core to be defined, from which other flow properties such as temperature, pressure, Mach number, and velocity, could be found from isentropic relations. Fig. 5 shows the necessary outputs from the plume model are ρ , T , γ , and u . γ is assumed constant within the entire plume, and ρ is shown above to decay throughout the plume. The change of T and u is more complex. As discussed in Section II, the breakdown of thermal equilibrium of the flow leads to a freezing (minimum) temperature and limiting (maximum) velocity. $P = 2$ was used to define the freezing surface in this work, at which the plume was assumed free molecular, and $u = u_{lim}$ and $T = T_f$ fixed. u_{lim} is constant within the plume core, whereas T_f decays with ϕ . Legge's proposal for effective stagnation conditions was used for the boundary layer: ρ_0 , p_0 , and T_0 were assumed to decay exponentially with ϕ (when $\phi > \phi_0$), to account for viscous effects. Consequently, both u_{lim} and T_f are functions ϕ in the boundary layer.

III.iii. Plume Impingement Model

The gas-surface interaction between the plume and a debris surface causes a transfer of momentum and energy. The degree to which this transfer takes place for a particular collision depends on the accommodation coefficients introduced in Section II: momentum coefficients α_n and α_t , and thermal (energy) coefficient α_e . Schaff highlighted the implicit assumption that α_e describes energy transfer from all molecular degrees of freedom (translational, rotational, vibrational) evenly [42], and that this was found to be approximately true for translational and rotational components. However, the vibrational energy was altered to a lesser extent during observed impingement [49]. The temperature required to excite the fundamental frequencies ω_e of NH_3 and N_2 , defined as $\omega_e = 3336 cm^{-1}$ and $2359 cm^{-1}$ respectively [50, 51], was then estimated by $T_{vib} = \hbar \omega_e / k_B$ to be 4792K and 3392K for the two species. From Table 2, the maximum gas temperature considered in this analysis was 1200K. Hence, the energy contained within vibrational modes of NH_3 and N_2 was small within the considered plume flow, minimising errors in α_e relating to vibrational energy transfer.

Considering the rotational energy modes: rotational constants of 9.4443cm^{-1} and 1.9982cm^{-1} for NH_3 and N_2 respectively were used, yielding activation temperatures $T_{rot} = B/k_B$ of 13.58K and 2.87K for the two gas species [52]. Hence, rotational modes could not be considered frozen. As a result, the assumption of hyperthermal flow used in Ref. [25] has been dropped in this analysis, in favour of Schaaf and Chambre's momentum transfer model:

$$p = \frac{dF_n}{dA} = \frac{1}{2}\rho u^2 \frac{1}{S^2} \left[\left(\frac{2 - \alpha_n}{\sqrt{\pi}} S \sin \theta + \frac{\alpha_n}{2} \sqrt{\frac{T_w}{T}} \right) e^{-(S \sin \theta)^2} + \left((2 - \alpha_n) \left((S \sin \theta)^2 + \frac{1}{2} \right) + \frac{\alpha_n}{2} \sqrt{\pi \frac{T_w}{T}} (S \sin \theta) \right) \cdot (1 + \text{erf}(S \sin \theta)) \right] \quad (8)$$

$$\tau = \frac{dF_t}{dA} = \frac{\alpha_t \rho u^2 \cos \theta}{2\sqrt{\pi}} \left(e^{-(S \sin \theta)^2} + \sqrt{\pi} (S \sin \theta) (1 + \text{erf}(S \sin \theta)) \right) \quad (9)$$

Where $\frac{dF}{dA}$ has been split into its normal and tangential components, which represent meaningful values of pressure p and shear stress τ , that vary with the incidence angle of the surface θ relative to the plume centreline. Eqs. 9 and 8 are valid only in the free molecular regime, where the velocity of the plume is considered to follow a maxwellian distribution. The same assumption is used in the derivation of Schaaf and Chambre's energy transfer model:

$$\dot{q} = \alpha_e \rho \mathcal{R} T \sqrt{\frac{\mathcal{R} T}{2\pi}} \left(\left[S^2 + \frac{\gamma}{\gamma - 1} - \frac{\gamma + 1}{2(\gamma - 1)} \frac{T_w}{T} \right] \cdot \left[e^{-(S \sin \theta)^2} + \sqrt{\pi} (S \sin \theta) \{1 + \text{erf}(S \sin \theta)\} \right] - \frac{1}{2} e^{-(S \sin \theta)^2} \right) \quad (10)$$

Where speed ratio S represents the ratio of the local flow velocity u , to the most probable gas speed as given by a maxwellian distribution, and is defined in Eq. 11. All accommodation coefficients were taken as 0.95.

$$S = \frac{u}{\sqrt{2\mathcal{R}T}} \quad (11)$$

III.iv. Example Detumbling Impingement Sequence

The general results for heat flux and stress from above, were applied to an example detumbling sequence. Fig. 6 shows the configuration of the sequence, in which the plume impinges on a solar panel surface of a debris object. It was assumed that the debris object was tumbling initially at $5^\circ/s$, and impinged until the angular momentum was brought to zero. Impingement of the surface was modelled to only occur when $20^\circ \leq \theta \leq 160^\circ$. It was also assumed that for all surface incidence angles θ , the plume centreline was aligned with the centre of the surface.

The composition of the solar panel was simplified in this analysis. Fig. 5 shows that the focus of the analysis was on the surface layer of the solar panel, and an adjacent adhesive layer. Fig. 7 shows the through-thickness panel composition thus assumed in this analysis. The glass surface layer was assumed to be the only side of the solar panel to be impinged. An adhesive layer, and a high-emissivity coating layer were both modelled as zero-thickness layers. Consequently, both layers maintain thermal equilibrium with the glass surface layer, and provided no additional thermal capacitance to the surface. The panel was discretised using a lumped parameter thermal network (LPTN) model. The LPTN formulation and solver routine described in this section are based on the methods presented by Gilmore [53]. The surface mesh used was a cartesian grid of isothermal mesh elements, and each mesh element was lumped to a single representative node at the element centre. Conduction was modelled between surface nodes, and it was assumed that the surface was conductively isolated from all other parts of the debris. All nodes were modelled to radiate from their high-emissivity surface side, to a deep space node. The resultant thermal network for an example node m is shown in Fig 8.

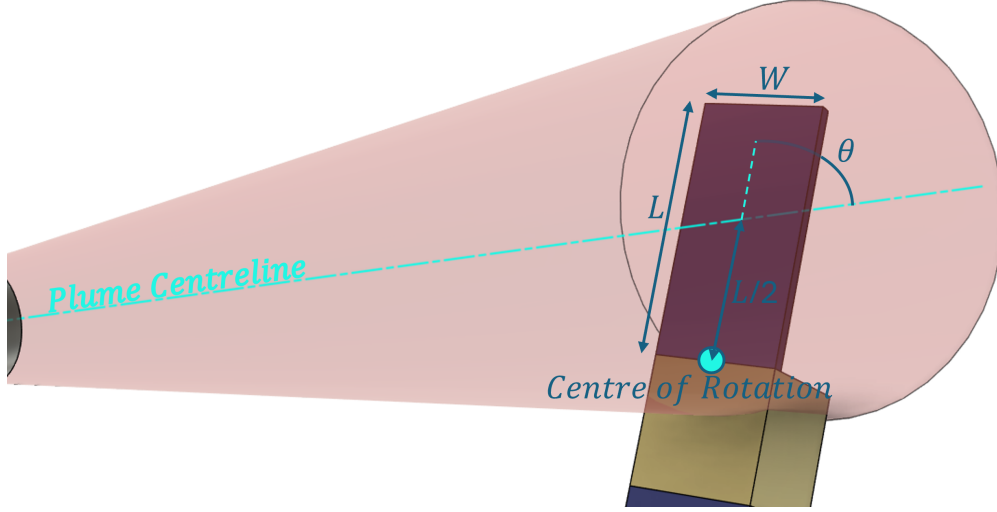


Fig. 6 Tumbling of an example debris object through incidence angle θ .

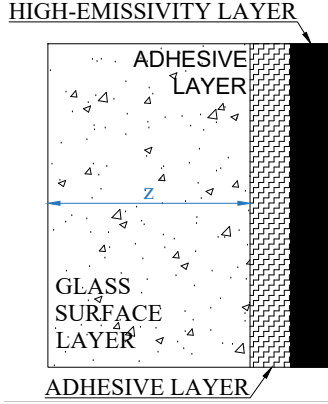


Fig. 7 Through-thickness composition of impinging surface.

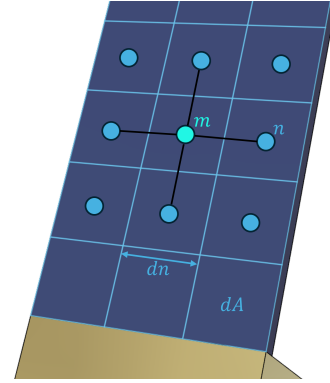


Fig. 8 Lumped parameter network example for a single node m to adjacent nodes n .

A further simplification to the radiation modelling was made by assuming that the surface was in equilibrium just before impingement. Normally in LEO, the solar heat flux, and heat flux from Earth infrared emissions and albedo, all change during an orbital period [53]. Instead, it was assumed that these heat fluxes were constant during the impingement sequence such that the surface reached thermal balance at a temperature of $T_w = 300K$. Hence, nodes were modelled to radiate to a representative isothermal space node at $T_{space} = 300K$, to satisfy the thermal balance requirement.

An explicit Euler finite forward difference time-stepping scheme was used to compute the temperature change per time step Δt , and is shown for a surface node m and adjacent nodes n in Eq. 12. The thermal capacitance of each node was calculated as $C_m = \rho_m dA_m z_m c_p$. Similarly, the Conductance K_{mn} between nodes m and n is defined as $K_{mn} = k_m \frac{dA_m}{dn_{mn}}$.

$$T_m^{j+1} = T_m^j + \frac{\Delta t}{C_m} \left(\sum_n K_{mn} (T_n^j - T_m^j) + Q_m^\theta - \sigma_B \epsilon_m dA_m \left((T_m^j)^4 - T_{space}^4 \right) \right) \quad (12)$$

Eq. 12 shows the temperature change to be dependent on both material and geometric properties. A surface $1m$ in width W and $2m$ in length L was considered in this analysis, discretised into mesh elements $0.2m$ in height and width.

Hence, $dn = 0.2m$, and $dA = 0.04m$. An emissivity $\epsilon = 0.9$ was assumed, and a range of thicknesses z were considered. Generalised material properties of three types of glass used in space solar panels are shown in Table 3; properties have been collated from Refs. [53] and [54]. $\dot{q}_m^\theta$ was calculated at each time step for each node. θ at time step j was evaluated based on the debris spin rate ω , such that $\theta^j = \theta^{j-1} + \Delta t \omega^j$. The spin rate was assumed to decrease linearly from $5^\circ/s$ at the start of the detumbling sequence, to $0^\circ/s$ after 1800 seconds, based on the results in Ref. [25].

Table 3 Material Properties of glasses considered in this analysis.

Glass Type	k [W/mK]	c_p [J/kgK]	C_{TE} [$\mu strain/K$]	K_C [MPa $\sqrt{m}$]	E [GPa]
Borosilicate Glass	1.12	830	3.25	0.60	64
Fused Silica	1.38	750	0.50	0.65	72
Quartz	1.35	750	0.45	0.67	73

The determination of temperature allowed the two debris creation scenarios of Fig. 5 to be studied. Analysing the risk of adhesive melting was achieved simply by comparing the peak temperature reached during the detumble sequence T_{peak} , to generalised melting temperatures of different classes of adhesives T_{crit} , to see if $T_{peak} < T_{crit}$. For the scenario of microcrack propagation, the peak stress incident on the surface σ_{peak} was found, considering p , τ , as well as the stresses induced by thermal expansion of the surface. Here, the stress from thermal expansion was found as $\sigma_e = (T_{peak} - 300)EC_{TE}$, and it was assumed that the surface was not allowed to expand freely. Determination of the critical stress was found using Irwin's theory [55]:

$$\sigma_{crit} = \sqrt{\frac{EG_c}{\pi a}} \quad (13)$$

The fracture energy of the material G_c is related to the fracture toughness. In this detumbling sequence, the surface was thin compared to its length and width, and hence it was assumed to be in plane stress. From this assumption, G_c was defined as:

$$G_c = \frac{K_c^2}{E} \quad (14)$$

And hence σ_{crit} could be evaluated and compared to σ_{peak} for the different glasses from Table 3, and for a range of assumed initial microcrack lengths a .

IV. Results

Table 4 shows nozzle flow properties at the throat and exit plane, calculated using CEA. The result $\gamma_E = 1.13$ differs considerably from experimental calculations of γ for hydrazine thrusters, including for a 5N thruster [36]. As a result, $\gamma = 1.37$ was used in subsequent analysis, as was reported in Ref. [36] and used in Refs. [25, 34]. Table 4 also shows the mass fraction of H_2 to be low at the exit plane and hence in the plume flow. Therefore, H_2 was not considered in subsequent calculations of Kn . The full output file from the CEA analysis is included in Appendix.

Table 4 Nozzle exit plane parameters, found from CEA analysis.

Parameter	Value in Chamber	Value at Throat	Value at Exit Plane
Mach Number [-]	0.000	1.000	5.334
Temperature [K]	1200.00	1017.58	317.79
Pressure [Pa]	1,100,000	587,780	505.38
Density [kg/m ³]	1.1784	0.74279	2.4913e-3
Sonic velocity [m/s]	1123.4	1038.7	479.73
Ratio of Specific Heats [-]	1.3520	1.3634	1.1345
Gas mixture molecular weight [kg/mol]	0.013026	0.013026	0.013026
Gas mixture viscosity [Ns/m ²]	4.1492e-5	3.7206e-5	1.5076e-5
Mass Fraction - H_2 [%]	12.56	12.55	7.45
Mass Fraction - NH_3 [%]	0.11	0.15	28.70
Mass Fraction - N_2 [%]	87.33	87.29	63.82

IV.i. Plume

The values of Table 4 were used as input to Simons' model. Fig. 9 shows contours of constant density, Mach number, and Bird's parameter. The contours are shown in cartesian coordinates where x corresponds to the nozzle centreline axis. The limiting and boundary layer angles are also shown. A clear distinction in the rate of decay between the isentropic core and the viscous boundary layer is visible in the contours of M . Importantly, M as it is presented in Fig. 9, is formulated as a function of density as opposed to velocity and temperature. This has been done to aid comparison to the results of Ref. [37]. Were $M(u, T)$ to be calculated instead, it would reach a limiting value at the freezing surface.

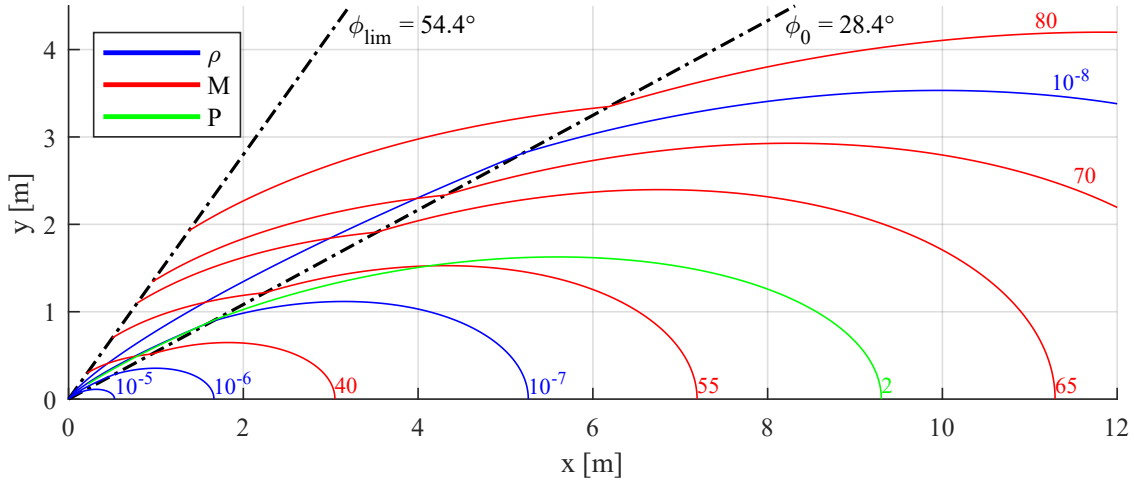


Fig. 9 Plume model generated with Simons' model, showing decay of density, Mach number and the freezing surface at $P = 2$. x is aligned with the plume centreline.

The contour of $P = 2$ in Fig. 9 was used to define the freezing surface. Along the centreline ($\phi = 0^\circ$), freezing occurs at $r = 9.30m$; when $\phi = \phi_0$, freezing occurs when $r = 1.92m$. Evidently, for $d = 10m$, a debris object would be within the free molecular plume region for all angles of $\phi \leq \phi_{lim}$. Fig. 10 shows the increase in Knudsen number with r for $\phi = 0^\circ$. For both reactant products, and for the range of L identified in Section I, $Kn > 10$ when $r = 10m$. Hence, the free molecular model of impingement is also correct to use for $d > 10m$.

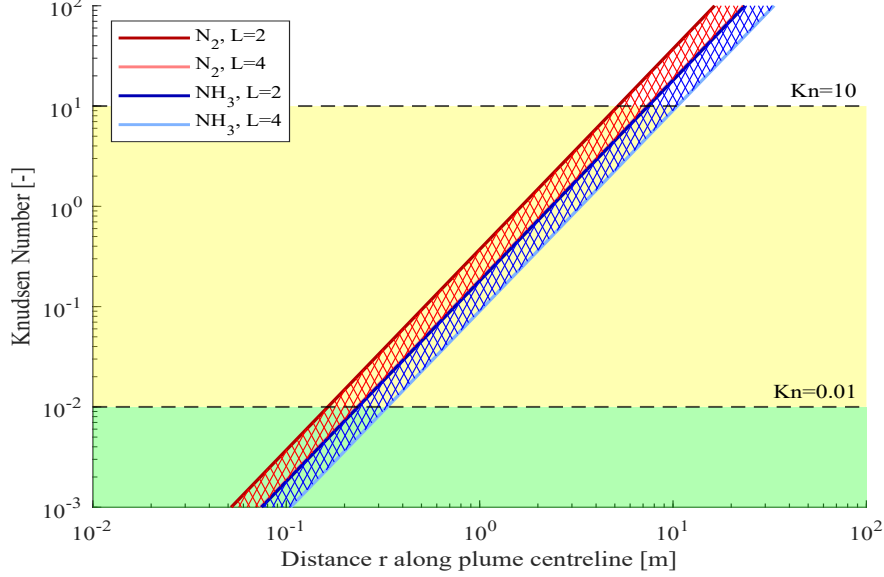


Fig. 10 Knudsen number increase along plume centreline, for different characteristic lengths L .

Fig. 11 shows that u and T approach close to a limiting value by $r = 1m$, and almost exactly by $r = 10m$. Hence, $P = 2$ appears to be a valid freezing surface, after which u and T_f are fixed at u_{lim} and T_f .

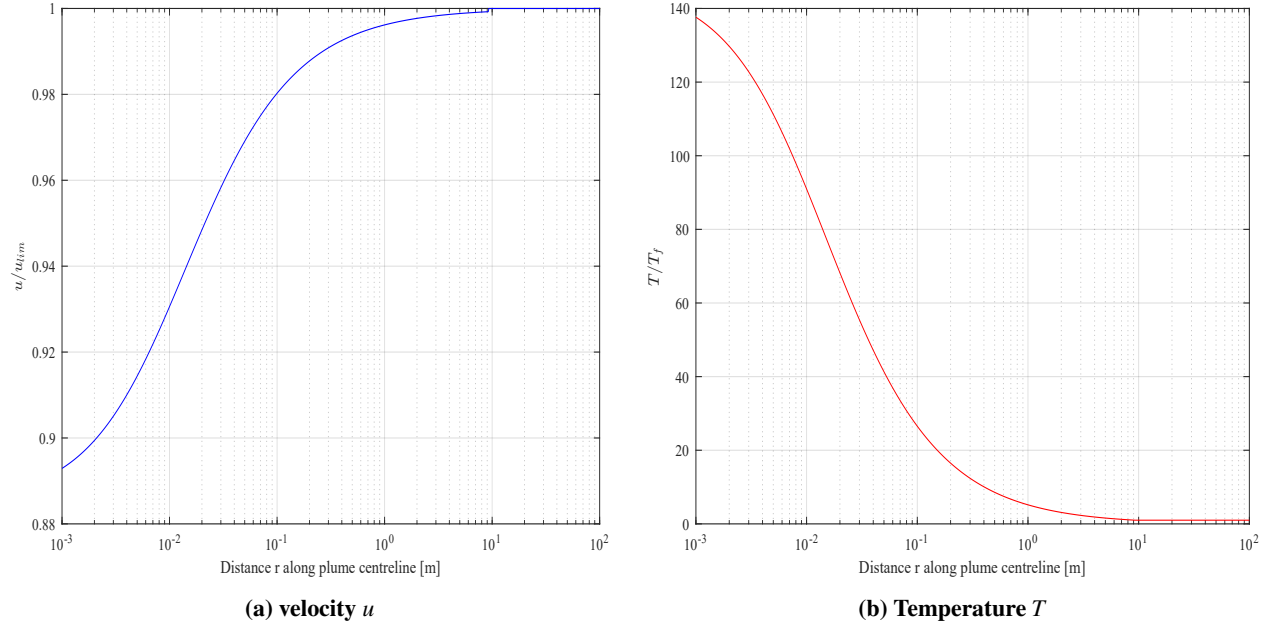


Fig. 11 Variation of velocity and temperature along plume centreline, relative to their freezing values.

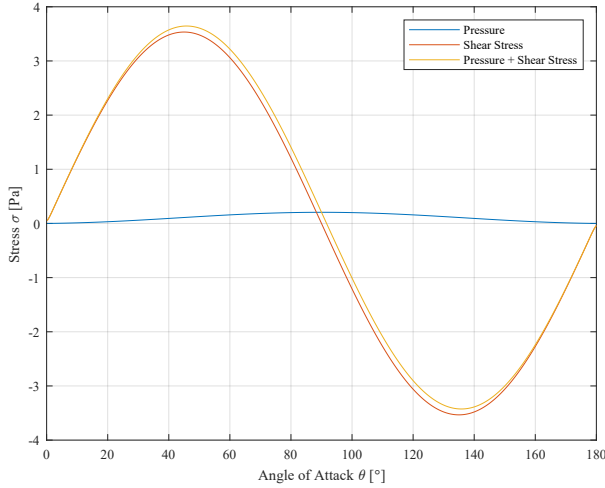
Plume constants, exit plane parameters, and flow properties at $r = 10$ and $\phi = 0^\circ$, are also reported. M was again calculated as a function of density. However, since $(r = 10m, \phi = 0^\circ)$ is close to the freezing surface, M is underpredicted by only 2.55% when calculated as $M(u, T)$. In both cases the flow is still found to be hypersonic.

Table 5 Parameters calculated by Simons' model

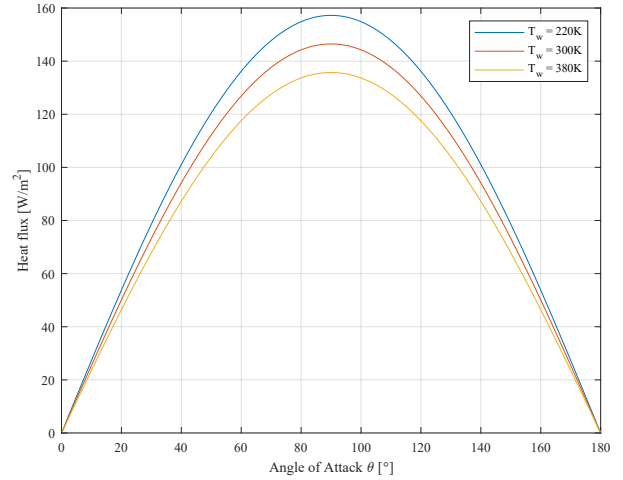
(a) Constants and at nozzle exit plane		(b) At $r = 10m, \phi = 0^\circ$	
Parameter	Value	Flow Properties	Value
Plume Constant A_P [-]	5.334	Density ρ [kg/m ³]	2.762e-8
Decay Constant c_ρ [-]	0.832	Velocity u [m/s]	2382
Boundary Layer Thickness δ_E [mm]	0.876	Temperature T [K]	1.768
Limiting velocity u_{lim} [m/s]	2382	Mach Number M [-]	62.17
Reynold's Number Re_E [-]	1184	Bird's Parameter P [-]	2.299
Frozen Temperature T_f [K]	1.768	Knudsen Number Kn [-]	18.05

IV.ii. Impingement

Fig. 12a shows the impingement results on a surface node at $r = 10m$ and $\phi = 0^\circ$, during half a rotation of the debris object. Maximum pressure occurs at $\theta = 90^\circ$, and maximum shear stress at $\theta = 45^\circ$. τ_{max} is 17 times greater than p_{max} , and hence the peak stress from momentum transfer $\sigma_{max} = 3.64Pa$ was found at $\theta = 45.9^\circ$. Impinged heat flux is dependent on T_w , and so Fig. 12b is presented for a range of T_w comparable to typical temperature ranges experienced by satellites in LEO [53]. Maximum heat fluxes of 157.2, 146.5, and 135.7W/m², occur for $T_w = 220, 300$, and 380K respectively. In all cases the maximum heat flux is at $\theta = 90^\circ$.



(a) Pressure and Shear Stress due to momentum transfer



(b) Heat flux due to energy transfer

Fig. 12 Impingement onto a Debris surface node at $r = 10m$, $\phi = 0^\circ$.

Fig. 13 shows the temperature increase over time for a surface node initially at 300K, with $\theta = 90^\circ$ and $\theta = 20^\circ$. These incidence angles represent the peak and minimum heat flux cases, for a detumbling scenario where the surface is only impinged when $20^\circ \leq \theta \leq 160^\circ$. For $\theta = 90^\circ$, thermal balance is reached at 313.78K; when $\theta = 20^\circ$, balance occurs at 305.12K. In both cases the rate of temperature change depends strongly on surface thickness. The minimum thickness of 0.03mm is the smallest thickness available of *Schott Solar Glass 0787* [56]. In this case, a maximum temperature was reached after 45 seconds. For a surface thickness of 50mm, temperature increases is close to linear with time, and

increases at approximately $0.0014K/s$.

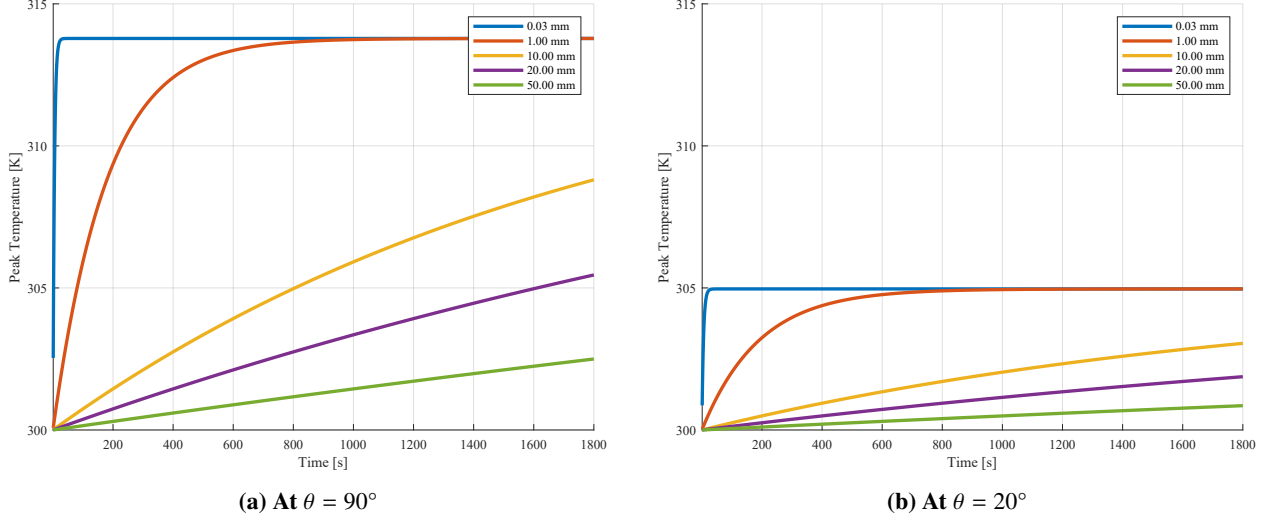


Fig. 13 Peak temperature of surface node at $r = 10m$, $\phi = 0^\circ$ during continuous impingement, for a range of surface thicknesses

When modelling the temperature change of a surface, heat fluxes due to impingement, conduction, and radiation were considered as shown in Section III. The temperature of T_{space} was taken to be that of $T_w = 300K$ in the case of Fig. 13. This gives the assumption that the surface is in thermal balance at 300K, and hence gave an indication of the temperature change induced by the plume impingement. The same value of T_{space} was used when simulating a more realistic detumbling scenario. In it, the results of which are shown in Fig. 14. Impingement of a 1mm thick surface was assumed to occur between $20^\circ < \theta < 160^\circ$, and the spin rate of the surface to decrease linearly during impingement phases. It was assumed that a total of 1800 seconds of impingement was required to detumble the debris object, from an initial spin rate of $5^\circ/s$. The peak temperature reached in the manoeuvre is $312.25K$ ($1.53K$ less than the continuous case of Fig. 13a). What is also clear from Fig. 14 is that ΔT increases as spin rate decreases- due to longer impingement periods.

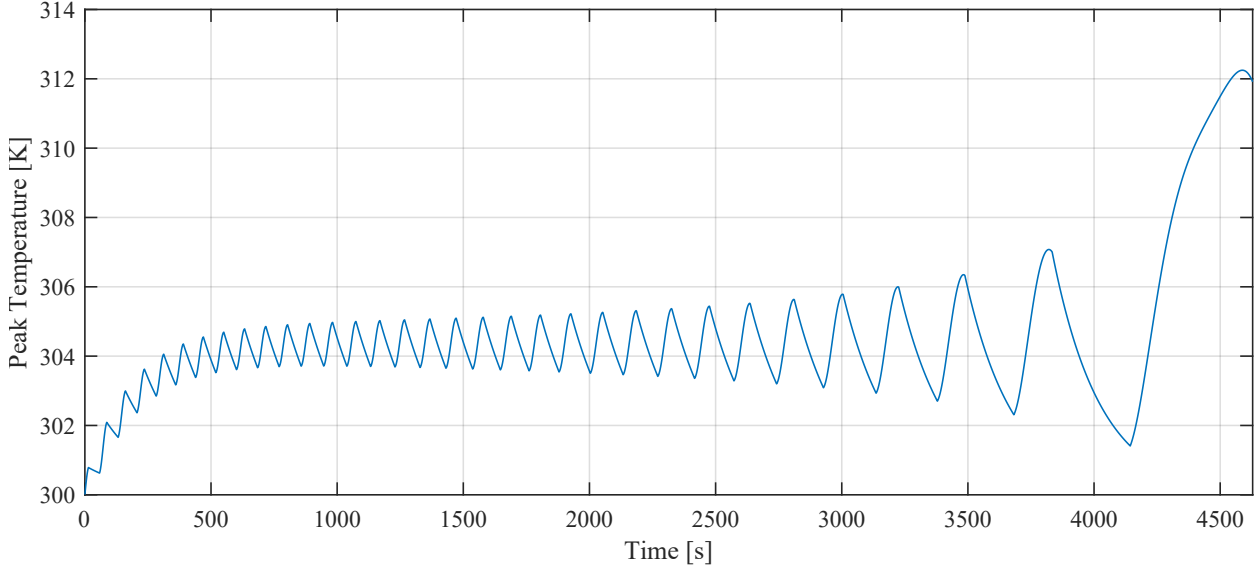


Fig. 14 ΔT for 1mm thick surface during a detumbling scenario from $5^\circ/s$; impingement when $20^\circ < \theta < 160^\circ$

The results of Fig. 14 were used to calculate the peak thermal stress induced in the surface, according to Eq. X. This thermal stress was then summed with the peak stress found from Fig. 12a. These peak stresses are shown in Fig. 15, compared to the critical fracture stress for a range of initial crack lengths. This comparison is given for three types of glass; relevant material properties for these glasses are shown in Table 3. It can be seen that the induced stresses will not cause fracture propagation in any of the glass types, for initial cracks of 10mm or smaller. Both fused and quartz glasses have similar peak induced stresses, whereas the safety margin for borosilicate glass is noticeably smaller. What is clear in all cases is that the stress from momentum transfer is exceedingly small compared to induced thermal stresses.

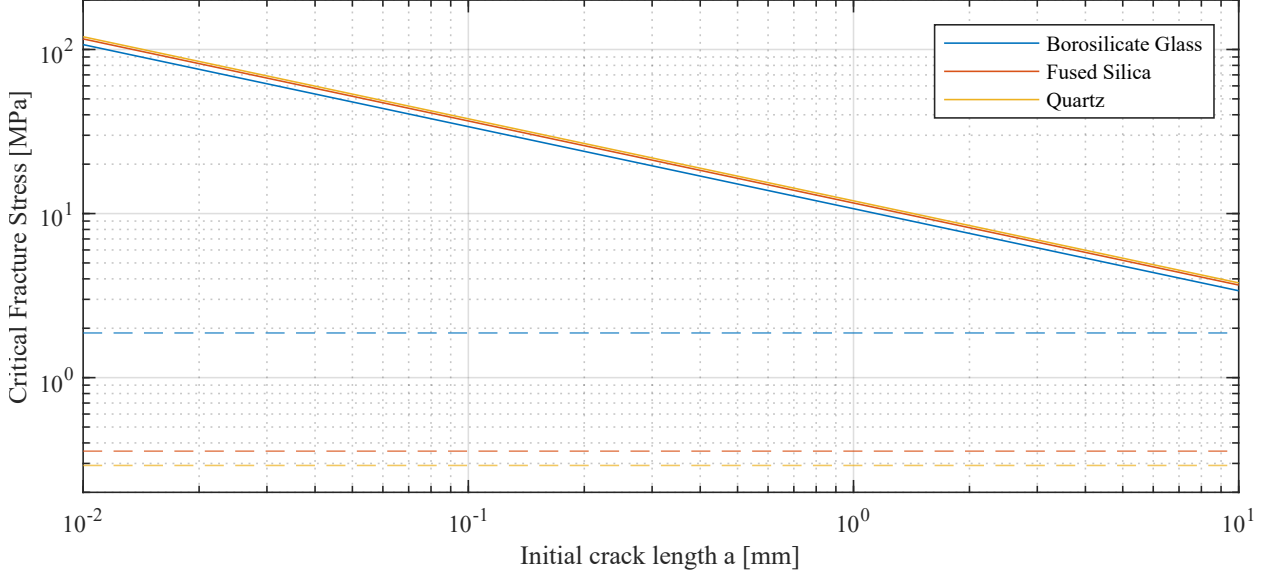


Fig. 15 Critical fracture stress compared to peak impinged stress, for three glass types.

V. Discussion

The results presented in Section IV show the effects of energy transfer to be much greater than momentum transfer. However, neither leading to significant risk of debris breakup for the two evaluated scenarios during the example detumbling scenario. The results are discussed below, in particular the validity and limitations of Simons' model in the free molecular regime, and the effect on the presented results.

Considering first the impingement results, Fig. 12a shows the maximum impinged stress due to momentum to be approximately $2Pa$. Peak shear stresses are approximately ten times the pressure forces, a result confirmed by other reported results using Schaaf and Chambre's method to calculate drag and lift for flat plate [57]. Nevertheless, both p and τ are still small. $2Pa$ is approximately 10^5 times greater than stresses resultant from solar radiation pressure and atmospheric drag at 500km orbit [53], but it is still well below typical glass yield strength exceeding 100MPa [54]. From Fig. 15, it is clear that stresses induced due to thermal expansion of the surface are much more significant: in the order of $0.3 - 1.1\text{MPa}$. It is to be expected that the effects of energy transfer are of a larger scale, compared to those of momentum transfer. As discussed in Section III, there is a transfer of translational and rotational energy of an impinging molecule. Considering only the translational energy, this kinetic energy is $\propto u^2$. On the other hand, momentum transfer is only proportional to u . In the free molecular region when $u_{lim} = 2382\text{m/s}$ is reached, this leads to energy transfer at least 10^3 times greater than momentum transfer. Fig. 15 shows that for the case of fracture propagation, the total impinged stress becomes close to the critical fracture stress for large initial crack lengths (particularly for borosilicate glass). However, at more common crack lengths of the μm scale, the safety margin for this case is still in the order of 10-100. A maximum temperature increase for a 1mm thick surface of 13K was reached for the continuous impingement presented in Fig. 13, and 12K for the detumbling sequence of Fig. 14. These temperature increases are less than typical transient temperature ranges experienced by external surfaces during an orbital period. Further, for the scenario

considered, the heat flux was an order of magnitude less than that typical from solar radiation in LEO. It is clear from this that there is no risk of causing adhesive melting, if that adhesive is rated for the spacecrafts operations. There is also a clear mitigation to further reduce the risk of debris breakup: where plume impingement be only performed during eclipse periods, such that the total heat flux imparted onto the surface is never greater than during an illuminated, non-impinged case.

Considering now the results of the plume model, Fig. 9 allows direct comparison to Legge's results, where excellent agreement is seen [37]. In general, there is a discrepancy of an order of magnitude in the density of Legge's model compared to that of Fig. 9, for example $\rho(r = 0.226m, \phi = 0^\circ)$ is 11 times smaller in Legge's result. This is consistent with the size of thruster studied: $0.5N$, $r^* = 0.3mm$ by Legge; and $5N$, $r^* = 1mm$ in this analysis. It follows from Eq. 4 that $\rho \propto (r^*)^2$, and hence an order of magnitude $((1/0.3)^2 = 11.11)$ difference in density suggests the plume model of Fig. 9 to be consistent with Legge's. Comparison with Boyd's results show a similar degree of agreement with the same unity order magnitude scaling [34]. Slightly larger discrepancies between this report's results and Boyd are seen compared to Legge's, likely due to the larger difference in stagnation conditions used in Boyd's work.

Rotation of the surface about the centre of rotation (Fig. 1) lead to a minimum distance of $r=9m$ being considered in this analysis. Fig. 9 shows this to be slightly before the freezing surface. Further, Fig. 10 shows that, for NH_3 , this is also before the free molecular limit defined by Kn . A minimum Knudsen number of $Kn \approx 7$ could occur at $r = 9m$. Bridging relations presented by Legge suggest that for $Kn = 7$, pressure coefficient and hence impinged pressure is increased by approximately 10% of $c_{p,c} - c_{p,FM}$ compared to the free molecular value. In the case of this analysis that is an increase in the order of 0.6 in impinged pressure. Legge's same bridging relations suggest shear stress coefficient is unchanged from its free molecular value at $Kn = 9$. Such small increased to impinged pressure and shear stress can be seen from Fig. 12a to be of little significance, in particular a 0.6 order increase in maximum shear stress would increase σ_{peak} by only 1.2Pa. Hence, modelling the entire impingement scenario as free molecular seems a reasonable simplification in this case.

Looking further at the plume model, analytical plume models of the form of Eq. 3 are known to cause underpredictions of plume temperature, due to the effects of flow condensation not captured in the flowfield [33]. Some mitigation of condensation effects was achieved in this work by using CEA with the equilibrium solver, which accounts for flow condensation in the nozzle - increasing the predicted exit temperature. Nevertheless, the plume model itself assumes no condensation. Results for a 0.5N thruster reported in Ref. [34] showed an underprediction of impinged pressure of $10^2 Pa$. Whilst more significant than the errors relating to the transition regime discussed above, even if p were to be underpredicted by $10^2 Pa$ this would still only cause an underprediction of σ_{peak} of 0.1% for the worst case of borosilicate glass. Consequently, the peak stresses experienced by the surface would still be below the critical stress for fracture propagation.

Another possible source of error in the plume model was the use of average gas properties of the reaction products, rather than modelling N_2 and NH_3 separately. Whilst the average property approach is generally followed in plume impingement studies using analytical models, it comes with the implicit assumption that the mass fractions of reactant products is constant throughout the plume. Experimental evidence gathered by Allen et al. found that in fact, product species of lower molecular mass tended to be pushed to greater angles of ϕ and into the boundary layer, and a concentration of heavier species was found close to the plume centreline [58]. This would have the effect of increasing density and subsequently p , τ , and $\dot{q}$ close to the plume centreline. The precise impact of this species concentration is not possible to determine without more involved numerical methods like DSMC.

VI. Future Work

It is clear that a lack of numerical or experimental results for direct comparison, limits the analytical model's results presented in this work. Future work should focus on the generation of comparable results using numerical or experimental methods. A hybrid CFD/DSMC approach seems the most suitable, like that of Nakajima [24]. In addition, DSMC simulations would allow the study of more complex geometries, where multiple particle reflections become important. In particular, greater refinement to the layer stack of a solar panel, may allow for better estimations of heat transfer across different layers (and materials) of a solar panel.

From results found in this work, it is clear that energy transfer and hence momentum flux are the major cause of any impingement effects. Hence, it makes sense for additional work to be done on improving the thermal solver used in this work. In particular, coupling of the plume model outputs of this work with more powerful thermal software suites like Airbus SYSTEMA and ESATAN could allow not only for the modelling of complex geometries, but also for the assumption of non-impingement thermal balance to be dropped. Instead, impingement could be modelled as part of a realistic orbital cycle of incident heat fluxes.

Of course, further work may be performed by considering more impingement scenarios. In particular, the consideration of impingement effects onto pressurised fuel tanks is a possible situation that warrants investigation. Finally, the merging of the method and models presented in this work, with models presented in work such as Peters and Olmos [25], would improve the modelling of the rotation rate of the impinged surface, as well as allow for less simplistic tumbling patterns to be simulated.

VII. Conclusions

The results of a plume impingement study, particularly the thermal and mechanical effect of an RCS thruster on a debris object, have been presented. Critically, they suggest the risk of debris breakup from plume impingement is low. For a distance of 10m between the thruster nozzle and impinged surface, peak stresses due to momentum transfer from the plume to the surface were found to be approximately $3.64Pa$. Energy transfer for the same condition caused peak heat fluxes of $146.5W/m^2$ for a surface temperature of $300K$. This value is 1-2 times less than typical solar fluxes in LEO, and hence it has been suggested that one possible way to mitigate against debris breakup would be to perform plume impingement only in eclipse periods, ensuring the condition during impingement is never worse (thermally) than that of an illuminated orbital period. Peak momentum transfer was found to occur for an incidence angle of 46° between the surface and plume centreline, whilst maximum heat flux occurred at an incidence angle of 90° . Two possible causes of debris breakup were considered. For an example detumbling scenario where a 1mm thick glass surface was impinged, the maximum temperature increase was $12K$, and the peak induced stress in the surface in the order of $1MPa$. Neither of these effects were found to cause debris breakup for the three glass materials considered.

However, the lack of numerical or experimental results able to directly verify the analytical modelling presented in this paper must limit confidence in the reported results. Underpredictions of heat flux and pressure are known to occur for the analytical Simons' plume model used in this work. Further, simplifications to the modelling of the plume gas mixture, and the assumption that the impinged surface will always be in the free molecular regime, both increase errors in the results. It has been shown that for many cases, these errors are insignificant compared to the safety margins found for the example scenarios. However, more complex modelling by use of numerical techniques such as DSMC would give greater estimation of the magnitude of these errors, and give greater confidence to the results in general.

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Appendix

Included on the next page is the input and output data of the CEA analysis used to generate the parameters quoted in this report. A digital copy of this data, as well as the MATLAB script used to construct the plume model and implement the thermal solver, are included in a GitHub repository of Ref. [59].

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NASA-GLENN CHEMICAL EQUILIBRIUM PROGRAM CEA2, FEBRUARY 5, 2004
 BY BONNIE MCBRIDE AND SANFORD GORDON
 REFS: NASA RP-1311, PART I, 1994 AND NASA RP-1311, PART II, 1996

CEA analysis performed on Fri 07-Mar-2025 08:24:06

Problem Type: "Rocket" (Infinite Area Combustor)

prob case=_____4066 ro equilibrium

Pressure (1 value):

p,bar= 11

Supersonic Area Ratio (3 values):

supar= 121, 135, 80

You selected the following reactants:

reac

name N2H4(L) wt%=100.0000 t,k= 579.350

You selected these options for output:

short version of output

output short

Proportions of any products will be expressed as Mass Fractions.

output massf

Heat will be expressed as siunits

output siunits

Transport properties calculated

output transport

Input prepared by this script:/var/www/sites/cearun.grc.nasa.gov/cgi-bin/CEARUN/prepareInputFile.cgi

IMPORTANT: The following line is the end of your CEA input file!
 end

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 159.5 PSIA

CASE = _____

NAME	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
N2H4(L)		1.000000	81585.966	579.350

O/F= 0.00000 %FUEL= 0.000000 R,EQ.RATIO= 0.000000 PHI,EQ.RATIO= 0.000000

	CHAMBER	THROAT	EXIT	EXIT	EXIT
Pinf/P	1.0000	1.8715	2176.37	2475.05	1335.18
P, BAR	11.000	5.8778	0.00505	0.00444	0.00824
T, K	1200.00	1017.58	317.79	314.64	330.42
RHO, KG/CU M	1.1784 0	7.4279-1	2.4917-3	2.2245-3	3.8289-3
H, KJ/KG	2545.97	2006.55	-728.06	-753.95	-625.99
U, KJ/KG	1612.52	1215.24	-930.91	-953.74	-841.16
G, KJ/KG	-17443.1	-14943.9	-6021.66	-5995.07	-6130.04
S, KJ/(KG)(K)	16.6576	16.6576	16.6576	16.6576	16.6576
M, (1/n)	10.689	10.692	13.026	13.094	12.768
(dLV/dLP)t	-1.00066	-1.00096	-1.11050	-1.11141	-1.10620
(dLV/dLT)p	1.0037	1.0062	2.9393	2.9720	2.8032
Cp, KJ/(KG)(K)	3.0021	2.9435	24.0731	24.4999	22.3278

GAMMAS	1.3520	1.3634	1.1345	1.1331	1.1402
SON VEL, M/SEC	1123.4	1038.7	479.7	475.8	495.3
MACH NUMBER	0.000	1.000	5.334	5.399	5.085

TRANSPORT PROPERTIES (GASES ONLY)

CONDUCTIVITY IN UNITS OF MILLIWATTS/(CM)(K)

VISC, MILLIPOISE	0.41492	0.37206	0.15075	0.14908	0.15735
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WITH EQUILIBRIUM REACTIONS

Cp, KJ/(KG)(K)	3.0021	2.9435	24.0731	24.4999	22.3278
CONDUCTIVITY	2.7026	2.3779	5.5829	5.6594	5.2692
PRANDTL NUMBER	0.4609	0.4606	0.6500	0.6454	0.6668

WITH FROZEN REACTIONS

Cp, KJ/(KG)(K)	2.9859	2.9118	2.3483	2.3374	2.3903
CONDUCTIVITY	2.6940	2.3631	0.7294	0.7158	0.7835
PRANDTL NUMBER	0.4599	0.4585	0.4853	0.4868	0.4800

PERFORMANCE PARAMETERS

Ae/At	1.0000	121.00	135.00	80.000
CSTAR, M/SEC	1425.8	1425.8	1425.8	1425.8
CF	0.7285	1.7948	1.8018	1.7666
Ivac, M/SEC	1800.5	2638.2	2646.8	2604.1
Isp, M/SEC	1038.7	2558.9	2569.0	2518.7

MASS FRACTIONS

*H2	0.12563	0.12554	0.07487	0.07366	0.07956
NH3	0.00106	0.00153	0.28695	0.29375	0.26054
*N2	0.87332	0.87293	0.63818	0.63259	0.65991

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS